Jem pays money into a savings scheme, *A*, over a period of 300 months.

Jem pays £20 into scheme *A* in month 1, £20.50 in month 2, £21 in month 3 and so on, so that the amounts Jem pays each month form an arithmetic sequence.

(a) Show that Jem pays £69.50 into scheme *A* in month 100

(1)

(b) Find the **total** amount that Jem pays into scheme *A* over the period of 300 months.

(2)

Kim pays money into a different savings scheme, *B*, over the same period of 300 months.

In a model, the amounts Kim pays into scheme *B* increase by the same percentage each month, so that the amounts Kim pays each month form a geometric sequence.

Given that Kim pays

- £20 into scheme *B* in month 1
- £250 into scheme *B* in month 300

(c) use the model to calculate, to the nearest £10, the difference between the total amount paid into scheme *A* and the total amount paid into scheme *B* over the period of 300 months.

(3)